CLASS 10 · CBSE · NCERT MATHEMATICS · SHORT NOTES

Chapter 10: Circles

Topic-wise short notes with solved examples

This chapter is about tangents to a circle: what a tangent is, how many can be drawn from a point, and the two theorems that every question is built on.

1 Lines and a circle

- A line and a circle in the same plane can meet in three ways.

Position of the line	Points in common	Name
Does not meet the circle	0	Non-intersecting line
Cuts the circle	2	Secant
Touches the circle	1	Tangent

- A **tangent** to a circle is a line that meets the circle at exactly one point. That point is the **point of contact**.
- A tangent is the limiting position of a secant when its two points of intersection come together.
- There is one and only one tangent at any given point of a circle.

2 Number of tangents from a point

Position of the point	Number of tangents
Inside the circle	0
On the circle	1
Outside the circle	2

- The **length of a tangent** from an external point P is the length of the segment from P to the point of contact.
- A circle has infinitely many tangents, one at each of its points. It has exactly two tangents parallel to a given line.

3 Theorem 1: tangent and radius

- **Theorem**: the tangent at any point of a circle is perpendicular to the radius through the point of contact.
- So if PT is a tangent at T to a circle with centre O, then $\angle OTP = 90^\circ$ and $\triangle OTP$ is right-angled at T.
- For a point P at distance d from the centre of a circle of radius r, the length of the tangent is

$$PT = \sqrt{d^2 - r^2}$$

- The perpendicular to a tangent at its point of contact passes through the centre.

Example 3.1 A point P is 13 cm from the centre of a circle of radius 5 cm. Find the length of the tangent from P to the circle.

SOLUTION

$OT \perp PT$, so $\triangle OTP$ is right-angled at T with $OP = 13$ cm and $OT = 5$ cm.

$PT = \sqrt{13^2 - 5^2} = \sqrt{144}$

$PT = 12$ **cm**

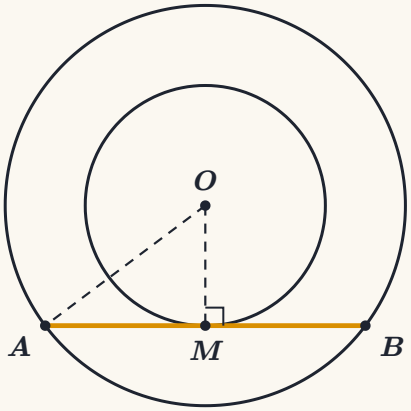
Example 3.2 The length of a tangent from a point A to a circle of radius 7 cm is 24 cm. Find the distance of A from the centre.

SOLUTION

$OA^2 = OT^2 + AT^2 = 7^2 + 24^2 = 49 + 576 = 625$

$OA = 25$ **cm**

Example 3.3 Two concentric circles have radii 13 cm and 5 cm. Find the length of the chord of the larger circle that touches the smaller circle.



Chord AB of the larger circle touches the smaller circle at M

SOLUTION

Let the chord AB touch the smaller circle at M. Then $OM \perp AB$, and the perpendicular from the centre bisects the chord, so $AM = MB$.

$AM = \sqrt{OA^2 - OM^2} = \sqrt{13^2 - 5^2} = 12$ cm

$AB = 2 \times 12 = 24$ **cm**

Example 3.4 Prove that the tangents drawn at the ends of a diameter of a circle are parallel.

SOLUTION

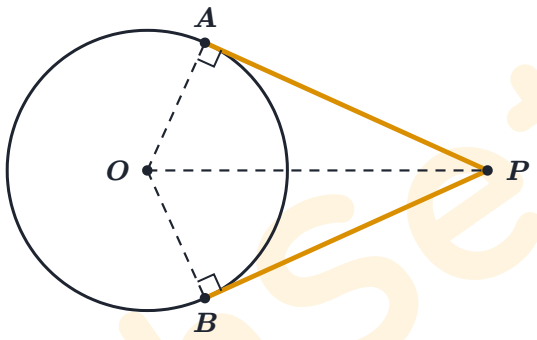
Let AB be a diameter and let the tangents at A and B be l and m.

$OA \perp l$ and $OB \perp m$ (tangent is perpendicular to the radius), so both l and m are perpendicular to the line AB.

Two lines perpendicular to the same line are parallel, so $l \parallel m$.

4 Theorem 2: tangents from an external point

- **Theorem**: the lengths of the tangents drawn from an external point to a circle are equal.



Tangents PA and PB from an external point P, each perpendicular to the radius at the point of contact

- If PA and PB are tangents from P to a circle with centre O, then $PA = PB$. The proof uses $\triangle OAP \cong \triangle OBP$ by RHS.
- From the same congruence: OP bisects $\angle APB$ and $\angle AOB$.
- $\angle APB + \angle AOB = 180°$, because the angles at A and B in quadrilateral $OAPB$ are both $90°$.

Example 4.1 PA and PB are tangents from P to a circle with centre O, and $\angle APB = 50°$. Find $\angle AOB$ and $\angle OAB$.

SOLUTION

$\angle AOB = 180° - \angle APB = 180° - 50° = 130°$

In $\triangle OAB$, $OA = OB$ (radii), so $\angle OAB = \angle OBA = \dfrac{180° - 130°}{2}$

$\angle AOB = 130°$ and $\angle OAB = 25°$

Example 4.2 A quadrilateral $ABCD$ is drawn to circumscribe a circle. If $AB = 8$ cm, $BC = 9$ cm and $CD = 6$ cm, find AD.

SOLUTION

Tangents from each vertex are equal, which gives $AB + CD = AD + BC$.

$8 + 6 = AD + 9$

$AD = 5$ **cm**

Example 4.3 From an external point P, tangents PA and PB are drawn to a circle. A third tangent at a point E of the circle meets PA at C and PB at D. If $PA = 12$ cm, find the perimeter of $\triangle PCD$.

SOLUTION

$CA = CE$ and $DB = DE$ (tangents from an external point)

Perimeter $= PC + CD + PD = PC + CE + ED + PD = (PC + CA) + (DB + PD) = PA + PB$

Perimeter $= 12 + 12 = 24$ **cm**

Example 4.4 The incircle of $\triangle ABC$ touches BC, CA and AB at D, E and F. If $AB = 12$ cm, $BC = 8$ cm and $CA = 10$ cm, find AF, BD and CE.

SOLUTION

Let $AF = AE = x$, $BD = BF = y$, $CD = CE = z$. Then $x + y = 12$, $y + z = 8$, $z + x = 10$.

Adding: $2(x + y + z) = 30$, so $x + y + z = 15$.

$AF = x = 15 - 8 = 7$ **cm**, $BD = y = 15 - 10 = 5$ **cm**, $CE = z = 15 - 12 = 3$ **cm**

5 Results worth remembering

- The centre of a circle lies on the bisector of the angle between two tangents drawn from an external point.
- A parallelogram circumscribing a circle is a rhombus.
- Opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre.
- In two concentric circles, a chord of the larger circle that touches the smaller circle is bisected at the point of contact.

6 Common mistakes

- Forgetting to state the reason "tangent is perpendicular to the radius at the point of contact" when marking a right angle.
- Taking the hypotenuse wrongly: in $\triangle OTP$ the hypotenuse is OP, the line from the centre to the external point.
- Assuming $\angle APB = \angle AOB$. They add up to 180°.
- Drawing a tangent from a point inside the circle. No tangent exists there.

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